

Deeksha Dadhich



PROFILE

Photonics engineer working across integrated-photonic design and characterisation, high-speed mixed-signal and RF/analog electronics, and system-level requirements definition. Experience spans modulator, switch, and resonator design; high-precision optical characterisation; and translating system-level performance targets into component specifications across optical, RF, electronic, thermal, and mechanical domains. Semiconductor background covering CMOS/VLSI design tools, high-speed DDR protocol and signal-integrity verification, and hands-on cryogenic hardware operation. Comfortable leading labs, managing engineers, and working across disciplines.

SKILLS

Photonic & EDA tools. Lumerical, IPKISS, GDSfactory, KLayout, COMSOL, Optsim, MATLAB, Cadence Virtuoso, Silvaco TCAD, Altium Designer, LTspice, NI LabVIEW.

Photonic device & switch design. Modulator and beam-splitter design; coupled microring-resonator design; PIC layout and test-structure design; >35 dB extinction-ratio optical switching; single-photon interferometry; tunable-laser systems.

Optoelectronic co-design & measurement. Optical, RF, and electronic requirements definition; high-precision insertion-loss, polarimetry, and extinction-ratio characterisation; Fock-space modelling linking system-level performance to component metrics; helium-pumped SNSPD installation & cryogenic operation; ARTIQ control.

Semiconductor & high-speed electronics. CMOS/VLSI design (Cadence Virtuoso, Silvaco TCAD); high-speed DDR2–DDR5 protocol and signal-integrity characterisation; JEDEC-compliant verification (SystemVerilog, UVM); mixed-signal integration.

Programming. Python (OOP, scientific stack, PyTorch), C/C++, SystemVerilog (UVM), Assembly.

Systems & hardware. PCB design, EMI/EMC analysis, CE certification, RF and analog electronics, mixed-signal integration, lab automation, test pipelines.

Leadership. Line management, vendor development, hiring, lab installation, cross-disciplinary requirements translation, technical mentoring.

PROFESSIONAL EXPERIENCE

Nu Quantum — Senior Photonics Engineer

Sep 2023 – Present

Quantum networking startup, Cambridge, UK

- Specified optical, RF-electronic, mechanical, and thermal requirements for the company's 4-node quantum entangler networking unit (QNU); led PIC characterisation across two flagship programmes (first generation fiber-based, second generation PIC-based).
- Designed companion PIC and test-bench circuits and defined PIC test structures during the design phase; developed Fock-space models linking system-level performance to component-level metrics.
- Developed precision measurement methodologies: >35 dB extinction-ratio switching, polarimetry, low-mean-photon quantum interferometry, and traceable high-precision insertion-loss characterisation.
- Emulated single-photon pulses from trapped-ion nodes on a representative test bench; installed and commissioned helium-pumped SNSPDs; built ARTIQ experiments for calibration and operation of the demo networking unit.
- Set budgets, timelines, and vendor requirements for new instrumentation; iterated specifications with equipment vendors; led purchasing decisions for the optics lab.
- Line-managed a senior optics engineer; lab manager for the optics lab; led installations of probe station and vacuum station; trained colleagues on PIC characterisation.
- Collaborated with Cambridge Consultants on commercialisation pathways for the QNU; read and translated state-of-the-art literature in quantum PIC design and measurement across disciplines.
- Through the Cornerstone *PIC Bootcamp*, designed and taped out silicon-photonic test structures — ring resonators and spiral waveguides for waveguide-loss extraction — laid out to the Cornerstone Silicon Photonics 1550 nm PDK in GDSfactory; tracked the die through fabrication (Cornerstone) and fibre packaging (Bay Photonics), and received the packaged, connectorised dies. Also completed AIM Photonics *Fabless PIC Design*.

ICFO — Institute of Photonic Sciences

2022, 3 months

Quantum Communications Research Intern, Barcelona

- Teamed with research group focused on opto-electronic quantum imaging and communication (Dr. Valerio Pruneri).
- Researched SPDC-based single-photon source characteristics for multi-core fiber quantum communication.
- Developed a free-space optical spectral analyzer with custom hardware and software for single-photon sources in visible and telecom bands.

Siemens EDA — Lead Engineer, DRAM IP Verification

2018 – 2021

Bengaluru, India

- Led development, testing, and support of latest DDR5 and DDR4 DRAM IP models for a major EDA verification product line.
- Implemented JEDEC-compliant protocol checking, assertions, coverage, and analysis components for memory controllers using SystemVerilog, C++, and UVM.
- Characterised DDR DRAM protocol performance (throughput, bus activity, power, timing) and built reusable SystemVerilog analysis components across DDR2–DDR5.

Grey Orange Robotics — Robotics R&D Engineer

2016 – 2018

Automated guided vehicle (AGV) electronic design team

- Designed PCBs and integrated systems for AGV warehouse robotics; owned product lifecycle planning and failure analysis.
- Led EMI/EMC testing, safety protocols, and CE certifications for AGV electronics shipped to multiple international sites.
- Designed a Power-over-Ethernet remote AGV controller integrating ARM M4, Zigbee radio, switch-mode power supply, sensors and actuators; deployed at multiple international sites and remained scalable for future additions.

Intel Labs — Optical Signal Processing Research Intern

2015, 6 months

Bengaluru, India

- Researched compressive-sensing signal-processing techniques to reduce power consumption in wearable optical sensors.
- Achieved ~70% power reduction for both sensor and communication subsystem using sparse-signal recovery.
- Co-authored US patent (P95539) for the sensing methodology and supported low-power hardware implementation.

Pai Kane — Power Engineering Intern

2013, 3 months

*Surveyed electric control panels***ACADEMIC PROFILE**

M.Sc. Photonics Engineering — Greatest Distinction

2021 – 2023

*Universiteit Gent & Vrije Universiteit Brussel***Junior Diploma — Hindustani Vocal Music**

2017 – 2019

*Prayag Sangeet Samiti***B.E. (Honors) Electronics & Instrumentation Engineering**

2012 – 2016

*Birla Institute of Technology and Science (BITS) Pilani***Industry courses***Fabless PIC Design (AIM Photonics); PIC Bootcamp (Cornerstone Foundry)***MOOCs & specialisations***Quantum Optics I & II (École Polytechnique); Laser Fundamentals & Applications (IIT Kharagpur); Semiconductor Optoelectronics (IIT Delhi); Active Optical Devices (CU Boulder)***Selected coursework***Electronics & ECE: Fiber Optics & Optoelectronics, Analog & Digital VLSI Design, Microelectronic Circuits, Electromagnetic Fields & Waves, Power Electronics, Electronic Instrumentation, Neural Networks & Fuzzy Logic.**Physics: Quantum Mechanics I, Special & General Theory of Relativity.***PUBLICATIONS & PATENTS**

Patent. *Compressive-sensing sparse-sampling photoplethysmogram (PPG) measurement.* V. Natarajan, D. Dadhich, K. Ranganathan. US P95539. Contributed the sparse-signal-recovery method demonstrating ~70% power reduction across sensor and communication subsystems.

Journal. *Design of a High-Gain Two-Stage CMOS Operational Amplifier.* Shruti, D. Dadhich, G. Meher. *IJIEASR*, Vol. 4 No. 6, June 2015.

SELECTED PROJECTS

Photonic chip microring resonator design

Universiteit Gent, Dr. W. Bogaert

- Designed, simulated, and characterised silicon photonic circuits for optical filtering applications.
- Used Python-based IPKISS to design coupled microring resonators; validated post-fabrication.

Integrated InP lasers for frequency shifting

Vrije Universiteit Brussel, Dr. M. Virts

- Researched integrated multi-mode lasers for signal-processing applications.
- Performed numerical and statistical modelling of laser nonlinear dynamics in MATLAB.
- Demonstrated a novel method for shifting optical signals by 1.3 THz with high-speed switching.

Characterising entangled photon sources for free-space quantum communication

ICFO, Dr. V. Pruneri

- Explored the spectral nature of SPDC-based entangled photon sources at different operating temperatures.
- Built a spectral analyzer combining APDs, coincidence counters, and wavelength filtering.

Ising machines for HPC with opto-electronics

Vrije Universiteit Brussel, Dr. G. Verschaffelt

- Implemented and benchmarked Ising-machine models in Python (time-to-solution, success rate).
- Demonstrated orders-of-magnitude improvement with a Chaotic Amplitude Control variant.

Linear Quantum Optical Computing Hackathon (Quandela, Paris)

Team of 5

- Reproduced quantum-photon speed-ups for reinforcement-learning methods using PERCEVAL simulators.
- Validated hybrid quantum-learning improvement over classical baselines; proposed enhancements.

Superresolution imaging with Convolutional Neural Networks

Vrije Universiteit Brussel, Dr. N. Deligiannis

- Designed a residual-learning CNN in PyTorch (NAdam optimizer) for $4\times$ superresolution.
- Outperformed interpolation baselines on the DIV2K benchmark.

GaN/InGaN nanowires for photoluminescence-based gas sensing

BITS Pilani, Dr. R. Singhal

- Studied InGaN nanowire UV absorption spectra in the presence of pollutant gases.
- Used COMSOL to model dependence on gas concentration, wire geometry, and spacing.

Neural-network-based heat-exchanger modelling

BITS Pilani, Dr. S. Bhanot

- Modelled the dynamic behaviour of a heat exchanger using artificial neural networks.
- Achieved high prediction accuracy on unseen experimental data.

SCHOLARSHIPS & AWARDS

VUB Master's Scholarship

2021 – 2023

Full two-year scholarship awarded to MSc students at VUB for motivation and excellence.

Rakesh Chandra Scholarship

2013

Awarded to one undergraduate per semester at BITS Pilani for academic and research performance.

Merit-cum-Need Scholarship

2012 – 2016

Awarded to students with meritorious performance in need of financial support.

COMMUNITY & VOLUNTEERING

Teaching children of lower socioeconomic privilege (Pilani, Delhi). Taught basic sciences, mathematics, and music using toys and simple experiments at the primary and middle-school level.

SPIE / OPTICA Student Chapter Brussels. Organised conferences and lectures with the BPhot student chapter.